PARISHRAM 2.0

CLASS 12TH 2026

HUMAN REPRODUCTION

ZOOLOGY | LECTURE-8

-PARAM SIR



TODAY'S TOPIC -

1. PARTURITION AND LACTATION
2. REVISION



PARTURATION

→ When muscle contraction starts, it lead to fetal ejection reflex called parturation

→ Gestation time ~ 9 month (± 7 days)

→ Neuroendocrine reflex action

- Nerve
- Hormone :- Oxytocin*

→ SIGNAL :-
1. Developed Foetus
2. Placenta

* Labor pain
- induce by Artifecial oxytocin
- PITOCIN
- Induce labour pain

EXTRACT

2.7 Parturition and Lactation

The average duration of human pregnancy is about 9 months which is called the gestation period. Vigorous contraction of the uterus at the end of pregnancy causes expulsion/delivery of the foetus. This process of delivery of the foetus (childbirth) is called **parturition**. Parturition is induced by a complex neuroendocrine mechanism. The signals for parturition originate from the fully developed foetus and the placenta which induce mild uterine contractions called **foetal ejection reflex.** This triggers release of oxytocin from the maternal pituitary. Oxytocin acts on the uterine muscle and causes stronger uterine contractions, which in turn stimulates further secretion of oxytocin. The stimulatory reflex between the uterine contraction and oxytocin secretion continues resulting in stronger and stronger contractions. This leads to expulsion of the baby out of the uterus through the birth canal – parturition. Soon after the infant is delivered, the placenta is also expelled out of the uterus. *What do you think the doctors inject to induce delivery?*

The mammary glands of the female undergo differentiation during pregnancy and starts producing milk towards the end of pregnancy by the process called **lactation**. This helps the mother in feeding the new-born. The milk produced during the initial few days of lactation is called **colostrum** which contains several antibodies absolutely essential to develop resistance for the new-born babies. Breast-feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby.

Child Birth

Signal ← Baby / Placenta

↓

Mild uterine Contraction (Labor pain)

↓

Foetal ejection Reflex

↓ TRIGGER

OXYTOCIN release from post. pituitary

↓

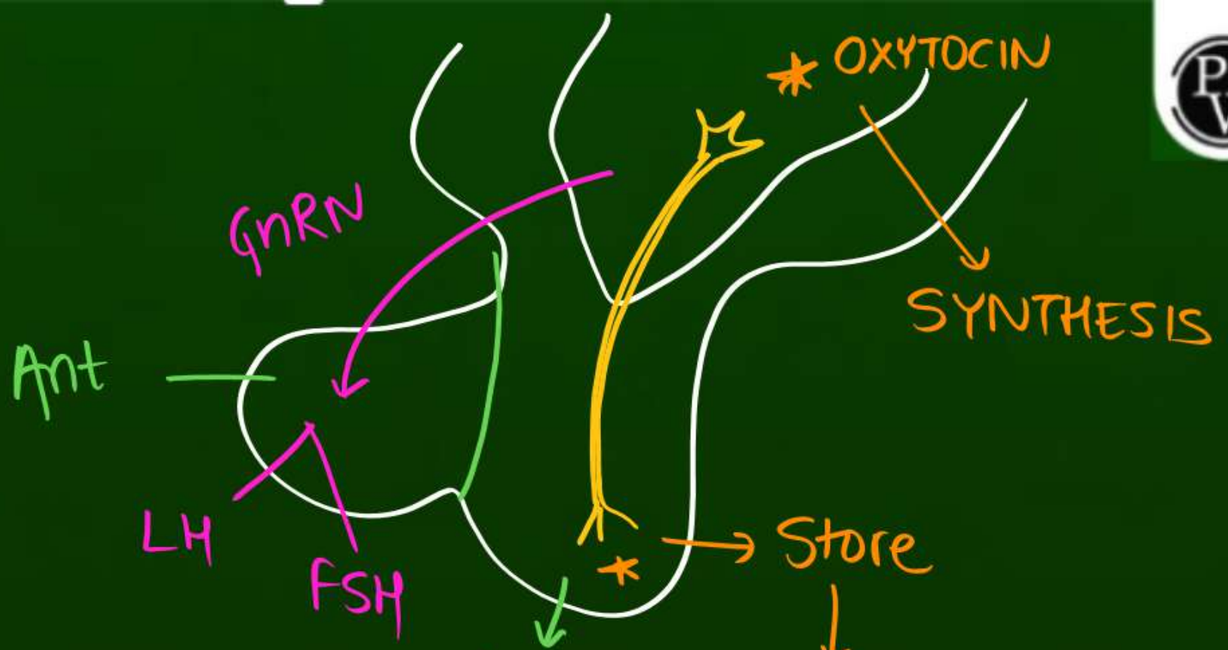
STRONG Uterine Contraction

↓

Expulsion of baby through BIRTH CANAL

Expulsion of Placenta

(+ve feedback loop)



→ PROLACTIN :- PRODUCTION
→ OXYTOCIN :- EJECTION

LACTATION

→ Mammary gland undergo differentiation at the time of Pregnency.

→ At parturation, start release of Milk Called Lactation

→ Colostrum : Initial milk of Mother
  - ↳ Antibodies (I_gA) : Protect from disease
  - ↳ Breast Feeding is recomended by doctor.

Homework

Colostrum is **(Pg-38)**

(a) Milk produced after parturition

(b) First milk

(c) Rich in antibodies and nutrients

(d) All of these

THANK YOU